



8544 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z14-0

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Jakob Helton (PI)	The Pennsylvania State University
Dr. George Rieke (CoI)	University of Arizona
Dr. Stacey Alberts (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Zihao Wu (CoI)	Harvard University
Dr. Daniel J. Eisenstein (CoI)	Harvard University
Dr. Kevin Hainline (CoI)	University of Arizona
Prof. Stefano Carniani (CoI) (ESA Member)	Scuola Normale Superiore, Pisa
Dr. Zhiyuan Ji (CoI)	University of Arizona
Dr. William Michael Baker (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Prof. Andrew Bunker (CoI) (ESA Member)	University of Oxford
Dr. Stephane Charlot (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Dr. Jacopo Chevallard (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Dr. Mirko Curti (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Emma Curtis-Lake (CoI) (ESA Member)	University of Hertfordshire
Dr. Francesco D'Eugenio (CoI) (ESA Member)	University of Cambridge, Kavli Institute for Cosmology
Dr. Eiichi Egami (CoI)	University of Arizona
Dr. Benjamin D. Johnson (CoI)	Harvard University
Dr. Nimisha Kumari (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Ms. Xiaojing Lin (CoI)	Tsinghua University
Dr. Jianwei Lyu (CoI)	University of Arizona
Prof. Roberto Maiolino (CoI) (ESA Member)	University of Cambridge

Name	Institution
Prof. Michael Maseda (CoI)	University of Wisconsin - Madison
Prof. Pablo G. Perez-Gonzalez (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Marcia J. Rieke (CoI)	University of Arizona
Dr. Pierluigi Rinaldi (CoI)	Space Telescope Science Institute
Prof. Brant Robertson (CoI)	University of California - Santa Cruz
Dr. Aayush Saxena (CoI) (ESA Member)	University of Oxford
Dr. Jan Scholtz (CoI) (ESA Member)	University of Cambridge, Kavli Institute for Cosmology
Dr. Fengwu Sun (CoI)	Harvard University
Dr. Sandro Tacchella (CoI) (ESA Member)	University of Cambridge
Dr. Giacomo Venturi (CoI) (ESA Member)	Scuola Normale Superiore, Pisa
Dr. Christina C Williams (CoI)	University of Arizona
Dr. Christopher Nicholas Andrew Willmer (CoI)	University of Arizona
Dr. Chris J. Willott (CoI) (CSA Member)	NRC Herzberg Institute of Astrophysics
Dr. Joris Witstok (CoI) (ESA Member)	Cosmic Dawn Center, Niels Bohr Institute
Dr. Yongda Zhu (CoI)	University of Arizona

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
MIRI/LRS Observations of JADES-GS-z14-0				
	1	MIRI/LRS Observations #1	MIRI Low Resolution Spectroscopy	(1) JADES-GS-z14-0
	2	MIRI/LRS Observations #2	MIRI Low Resolution Spectroscopy	(1) JADES-GS-z14-0
	3	MIRI/LRS Observations #3	MIRI Low Resolution Spectroscopy	(1) JADES-GS-z14-0
	4	MIRI/LRS Observations #1 (REPEAT)	MIRI Low Resolution Spectroscopy	(1) JADES-GS-z14-0

ABSTRACT

The recent spectroscopic confirmation of JADES-GS-z14-0 at $z = 14.179$ (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025) challenges our understanding of the early Universe. This galaxy is remarkably luminous ($M_{UV} = -20.81$) and reddened ($\beta_{UV} = -2.20$). It is clearly spatially resolved in the rest-frame ultraviolet ($R_{UV} = 260$ parsecs) and therefore dominated by stellar light at those wavelengths. Most notably, it

has been detected at 7.7 microns with JWST/MIRI, and the measured flux density in MIRI/F770W (74.4 nJy, S/N = 13) is in excess above the measured flux density with JWST/NIRCam (Helton et al. 2025). At this redshift, less than 300 million years after the Big Bang, the measured flux excess is likely caused by nebular emission from hydrogen and ionized oxygen. We propose for ultra-deep spectroscopic follow-up observations with the MIRI/LRS to cover the brightest rest-frame optical nebular emission lines ([OII]3727+3729, Hbeta, [OIII]4959+5007, and Halpha) in addition to the stellar continuum. These observations will unveil important insights into this extreme galaxy: (1) gas-phase metallicity, electron density, and ionization parameter; (2) recent star-formation rate, ionizing photon production efficiency, and diffuse dust attenuation; and (3) properties of the stellar populations, including stellar mass, stellar age, and shape of the star-formation history. JADES-GS-z14-0 represents the current archetype of early galaxy formation, and the proposed observations represent a rare opportunity to study the formation of some of the first stars and heavy elements in the Universe.

OBSERVING DESCRIPTION

We propose for ultra-deep spectroscopic follow-up observations of JADES-GS-z14-0 at $z = 14.179$ with the MIRI/LRS to cover the brightest rest-optical nebular emission lines ([OII]3727+3729, Hbeta, [OIII]4959+5007, and Halpha) in addition to the rest-optical continuum. We are requesting 62.8 hours of charged time including overheads (50.7 hours of science time). The proposed observations are a substantial investment for one galaxy, but we note that this proposal is building on nearly ten days of mission time spent observing JADES-GS-z14-0 with JWST, and that this galaxy is currently the ideal test bed for the study of early galaxy formation. Considering the extreme properties of JADES-GS-z14-0, opportunities like this one are expected to be rare. Furthermore, the existing photometric detection with JWST/MIRI (Helton et al. 2025) and spectroscopic detection with ALMA (Carniani et al. 2025, Schouws et al. 2024) provide fairly robust constraints on the expected rest-frame optical nebular emission line fluxes.

The proposed observations with the MIRI/LRS are for slit spectroscopy and adopt the FASTR1 readout pattern with 119 groups per integration, 23 integrations per exposure, 4 exposures per dither, 2 dithers along slit nod per visit, and 3 visits in total. We choose to have multiple visits to provide repeated dithering between both of the slit locations. For each visit, direct images are taken with the MIRI/F560W filter for target acquisition and verification. Verification is highly recommended for slit spectroscopy with the MIRI/LRS. Target acquisition would be performed on a bright, nearby star ($m = 21$ AB mag in MIRI/F560W) that is roughly 40 arcseconds to the north east. The proposed observations for target acquisition adopt the FASTGRPAVG readout pattern with 10 groups per integration and 1 integration per exposure, while the proposed observations for verification adopt the FASTR1 readout pattern with 119 groups per integration and 1 integration per exposure.

Proposal 8544 - Targets - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z14-0

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	JADES-GS-z14-0	RA: 03 32 19.9049 (53.0829371d) Dec: -27 51 20.28 (-27.85563d) Equinox: J2000	Epoch of Position: 2000	
		Comments: One of the most distant spectroscopically confirmed galaxies at $z = 14.179$: JADES-GS-z14-0 (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025). Category=Galaxy Description=[High-redshift galaxies] Extended=YES			
	(2)	Target-Acquisition-Star	RA: 03 32 22.8080 (53.0950333d) Dec: -27 50 53.65 (-27.84824d) Equinox: J2000	Proper Motion RA: 9.176 mas/yr Proper Motion Dec: 1.936 mas/yr Parallax: 0.0005808" Epoch of Position: 2000	
		Comments: Target acquisition star for JADES-GS-z14-0. Coordinates are from the JADES/NIRCam mosaics and photometric catalogs, which have all been registered to the Gaia DR3 frame. This star appears to be a K4/K5 giant based on the observed Gaia color ($BP - RP = 1.624308$ mag) and parallax (0.5808 mas with a standard error of 0.5918 mas). Alternate names for this source are EIS-DEEP CDFS-3 J 1 1119 and Gaia DR3 5057500088994710016. Category=Star Description=[K giants] Extended=NO			

Proposal 8544 - Observation 1 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Observation	Proposal 8544, Observation 1: MIRI/LRS Observations #1								Wed Nov 26 20:00:19 GMT 2025	
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	JADES-GS-z14-0	RA: 03 32 19.9049 (53.0829371d) Dec: -27 51 20.28 (-27.85563d) Equinox: J2000			Epoch of Position: 2000				
	Comments: One of the most distant spectroscopically confirmed galaxies at z = 14.179: JADES-GS-z14-0 (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025).									
	Category=Galaxy Description=[High-redshift galaxies] Extended=YES									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 Target-Acquisition-Star	F560W	FASTGRPAVG	10	1	1	111.002	258634	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	2	2	100.0		100.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	119	1	1	1	1	330.23		F560W

Proposal 8544 - Observation 1 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	119	23	46	1	2	15312.671	258634
Special Requirements	Aperture PA Range 339.75797 to 9.75797 Degrees (V3 335.0 to 5.0) Offset -0.11 arcsec, 0.0 arcsec Same Aperture PA 1, 2, 3								

Proposal 8544 - Observation 2 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Observation	Proposal 8544, Observation 2: MIRI/LRS Observations #2								Wed Nov 26 20:00:20 GMT 2025	
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	JADES-GS-z14-0	RA: 03 32 19.9049 (53.0829371d) Dec: -27 51 20.28 (-27.85563d) Equinox: J2000			Epoch of Position: 2000				
	Comments: One of the most distant spectroscopically confirmed galaxies at z = 14.179: JADES-GS-z14-0 (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025).									
	Category=Galaxy Description=[High-redshift galaxies] Extended=YES									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 Target-Acquisition-Star	F560W	FASTGRPAVG	10	1	1	111.002	258634	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	2	2	100.0		100.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	119	1	1	1	1	330.23		F560W

Proposal 8544 - Observation 2 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	119	23	46	1	2	15312.671	258634
Special Requirements	Offset 0.11 arcsec, 0.0 arcsec								
	Same Aperture PA 1, 2, 3								

Proposal 8544 - Observation 3 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Observation	Proposal 8544, Observation 3: MIRI/LRS Observations #3								Wed Nov 26 20:00:20 GMT 2025
	Diagnostic Status: Warning								
	Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous		
	(1)	JADES-GS-z14-0	RA: 03 32 19.9049 (53.0829371d)		Epoch of Position: 2000				
			Dec: -27 51 20.28 (-27.85563d)						
			Equinox: J2000						
	Comments: One of the most distant spectroscopically confirmed galaxies at z = 14.179: JADES-GS-z14-0 (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025). Category=Galaxy Description=[High-redshift galaxies] Extended=YES								
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	2 Target-Acquisition-Star	F560W	FASTGRPAVG	10	1	1	111.002	258634
Template	Subarray				Obtain Verification Image?				
	FULL				true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order	
	2	2	100.0		100.0	0.0	0.0	DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps	Spatial Step Offset
	1	ALONG SLIT NOD							
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID
	1	FASTR1	119	1	1	1	1	330.23	F560W

Proposal 8544 - Observation 3 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	119	23	46	1	2	15312.671	258634
Special Requirements	Offset 0.33 arcsec, 0.0 arcsec								
	Same Aperture PA 1, 2, 3								

Proposal 8544 - Observation 4 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Observation	Proposal 8544, Observation 4: MIRI/LRS Observations #1 (REPEAT)									Wed Nov 26 20:00:20 GMT 2025
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	JADES-GS-z14-0	RA: 03 32 19.9049 (53.0829371d) Dec: -27 51 20.28 (-27.85563d) Equinox: J2000			Epoch of Position: 2000				
	Comments: One of the most distant spectroscopically confirmed galaxies at z = 14.179: JADES-GS-z14-0 (Carniani et al. 2024, Carniani et al. 2025, Schouws et al. 2025).									
	Category=Galaxy Description=[High-redshift galaxies] Extended=YES									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 Target-Acquisition-Star	F560W	FASTGRPAVG	10	1	1	111.002	258634	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %	Row shift (deg)	Column shift (deg)		Tile Order	
	2	2	100.0		100.0	0.0	0.0		DEFAULT	
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
	1	FASTR1	119	1	1	1	1	330.23		F560W

Proposal 8544 - Observation 4 - Rest-Frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-Up for JADES-GS-z1...

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
	1	FASTR1	119	23	46	1	2	15312.671	258634
Special Requirements	Aperture PA Range 339.75797 to 49.75797 Degrees (V3 335.0 to 45.0) Offset -0.11 arcsec, 0.0 arcsec								